Assignment #2

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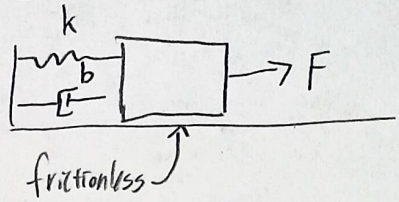
Given: Case study D, $m = 5$ kg, $k = 3$ N/m, $b = 0.5$ N·s/m

Find:
- Configuration variables, $\vec{q}$
- kinetic energy, $K(\vec{q}, \dot{\vec{q}})$

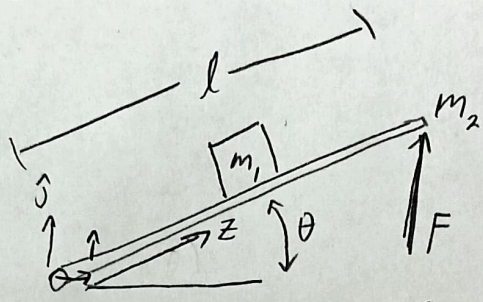


Solution:

- $\vec{q} = [z]$, $\dot{\vec{q}} = [\dot{z}]$ ← $\vec{q}$
- $K(\vec{q}, \dot{\vec{q}}) = \frac{1}{2} m \dot{z}^2 = \frac{1}{2} [m] \dot{\vec{q}}^T \dot{\vec{q}}$ ← $K(\dot{\vec{q}})$

Given:
- Case Study E
- $m_1 = 0.35$ kg
- $m_2 = 2$ kg
- $l = 0.5$ m
- $g = 9.81$ m/s²

Find:
- $\vec{q}$
- $K(\dot{\vec{q}})$



Assumptions:
- Neglect moment of inertia of m_1
- assume m_2 is slender rod

Solution:

- $\vec{q} = \begin{bmatrix} z \\ \theta \end{bmatrix}$, $\dot{\vec{q}} = \begin{bmatrix} \dot{z} \\ \dot{\theta} \end{bmatrix}$ ← $\vec{q}$

$$p_1 = \begin{bmatrix} z\cos\theta \\ z\sin\theta \\ 0 \end{bmatrix} \xrightarrow{\text{sympy}} \dot{p}_1 = \begin{bmatrix} -z\dot{\theta}\sin\theta + \dot{z}\cos\theta \\ z\dot{\theta}\cos\theta + \dot{z}\sin\theta \\ 0 \end{bmatrix}$$

$$p_2 = \begin{bmatrix} \frac{l}{2}\cos\theta \\ \frac{l}{2}\sin\theta \\ 0 \end{bmatrix} \Rightarrow \dot{p}_2 = \begin{bmatrix} -\frac{l}{2}\dot{\theta}\sin\theta \\ \frac{l}{2}\dot{\theta}\cos\theta \\ 0 \end{bmatrix} \qquad \omega = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix}$$

$$J_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{12} m_2 l^2 & 0 \\ 0 & 0 & \frac{1}{12} m_2 l^2 \end{bmatrix}$$

$$K = \frac{1}{2} m_1 \dot{p}_1^T \dot{p}_1 + \frac{1}{2} m_2 \dot{p}_2^T \dot{p}_2 + \frac{1}{2} \omega^T J_2 \omega \Rightarrow \text{sympy}$$

$$= \frac{1}{6} l^2 m_2 \dot{\theta}^2 + \frac{1}{2} m_1 (z^2 \dot{\theta}^2 + \dot{z}^2)$$ ← $K(\dot{\vec{q}})$

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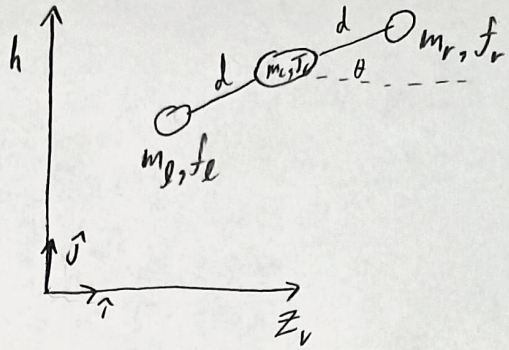
Given: · Case Study F

· $m_c, J_c, m_r, f_r, d, m_\ell, f_\ell$, etc.

Find: · $\vec{q}$

· $K(\vec{q})$



Assumptions: · m_ℓ and m_r point masses
· 2-D

Solution: Generalized coordinates...

$$\vec{q} = \begin{bmatrix} z_v \\ h \\ \theta \end{bmatrix} \Rightarrow \dot{\vec{q}} = \begin{bmatrix} \dot{z}_v \\ \dot{h} \\ \dot{\theta} \end{bmatrix} \longleftarrow \vec{q}$$

$$p_c = \begin{bmatrix} z_v \\ h \\ 0 \end{bmatrix} \Rightarrow \dot{p}_c = \begin{bmatrix} \dot{z}_v \\ \dot{h} \\ 0 \end{bmatrix}$$

positions & velocities...

$$p_r = p_c + \begin{bmatrix} +d\cos\theta \\ d\sin\theta \\ 0 \end{bmatrix} \Rightarrow \dot{p}_r = \begin{bmatrix} \dot{z} - d\dot{\theta}\sin\theta \\ \dot{h} + d\dot{\theta}\cos\theta \\ 0 \end{bmatrix}$$

$$p_\ell = p_c + \begin{bmatrix} -d\cos\theta \\ -d\sin\theta \\ 0 \end{bmatrix} \Rightarrow \dot{p}_\ell = \begin{bmatrix} \dot{z} + d\dot{\theta}\sin\theta \\ \dot{h} - d\dot{\theta}\cos\theta \\ 0 \end{bmatrix}$$

$$\omega = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix}, \quad J_c = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & J_c \end{bmatrix}$$

$$KE_c = \frac{1}{2} m_c \dot{p}_c^T \dot{p}_c + \frac{1}{2} \omega^T J_c \omega$$

$$KE_r = \frac{1}{2} m_r \dot{p}_r^T p_r$$

$$KE_\ell = \frac{1}{2} m_\ell \dot{p}_\ell^T p_\ell$$

$$KE = KE_c + KE_r + KE_\ell$$

Using sympy...

$$K(\vec{q}) = \frac{1}{2} J_c \dot{\theta}^2 + \frac{1}{2} m_c (\dot{h}^2 + \dot{z}_v^2) + \frac{1}{2} m_\ell \left((d\dot{\theta}\sin\theta + \dot{z}_v)^2 + (d\dot{\theta}\cos\theta - \dot{h})^2\right) + \frac{1}{2} m_r \left((d\dot{\theta}\sin\theta - \dot{z}_v)^2 + (d\dot{\theta}\cos\theta + \dot{h})^2\right) \longleftarrow K(\vec{q})$$